1 Control Overview

State space model of a robot ($\boldsymbol{x}_1 = \boldsymbol{q}$, $\boldsymbol{x}_2 = \dot{\boldsymbol{q}}$):

$$\dot{\boldsymbol{x}} = \begin{pmatrix} \boldsymbol{x}_2 \\ -M^{-1}(\boldsymbol{c}+\boldsymbol{g}) \end{pmatrix} + \begin{pmatrix} \boldsymbol{0} \\ M^{-1} \end{pmatrix} \boldsymbol{u} = f(\boldsymbol{x}) + G(\boldsymbol{x}_1)\boldsymbol{u} \tag{1}$$

Torque-current relation: $\tau_c = K_I i_c$; elastic joint torque: $\boldsymbol{\tau}_j = K(\boldsymbol{\theta} - \boldsymbol{q})$.

2 Lyapunov Stability

Equilibrium: $\boldsymbol{x}_e$ unforced if $f(\boldsymbol{x}_e) = 0$, $\boldsymbol{u} = 0 \Rightarrow \dot{\boldsymbol{q}} = 0$, $\boldsymbol{g} = 0$. Forced equilibrium: $\boldsymbol{u} = \boldsymbol{g}$.

Stability: $\forall \varepsilon > 0, \exists \delta_\varepsilon : \|\boldsymbol{x}(t_0) - \boldsymbol{x}_e\| < \delta_\varepsilon \Rightarrow \|\boldsymbol{x}(t) - \boldsymbol{x}_e\| < \varepsilon$

Asymptotic stability: $\exists \delta > 0 : \|\boldsymbol{x}(t_0) - \boldsymbol{x}_e\| < \delta \Rightarrow \lim_{t\to\infty} \|\boldsymbol{x}(t) - \boldsymbol{x}_e\| = 0$. GAS if $\forall \delta$ (unique equilibrium).

Exponential stability:

$$\|\boldsymbol{x}(t) - \boldsymbol{x}_e\| \le c\, e^{-\lambda(t-t_0)} \|\boldsymbol{x}(t_0) - \boldsymbol{x}_e\| \tag{2}$$

Lyapunov candidate: $V(\boldsymbol{x}_e) = 0$, $V(\boldsymbol{x}) > 0\ \forall \boldsymbol{x} \ne \boldsymbol{x}_e$.

$$\dot{V} = \frac{\partial V}{\partial \boldsymbol{x}} f(\boldsymbol{x}) \tag{3}$$

Thm (Lyapunov): $\dot{V} \le 0 \Rightarrow$ stable; $\dot{V} < 0$ (except at $\boldsymbol{x}_e$) $\Rightarrow$ asympt. stable; $\dot{V} > 0 \Rightarrow$ unstable.

Invariant set: $\boldsymbol{x}(t_0) \in \mathcal{M} \Rightarrow \boldsymbol{x}(t) \in \mathcal{M}\ \forall t \ge t_0$.

LaSalle: if $\exists V : \dot{V} \le 0$, system converges to largest invariant set $\mathcal{M} \subseteq S = \{\boldsymbol{x} : \dot{V}(\boldsymbol{x}) = 0\}$. If $\mathcal{M} = \{\boldsymbol{x}_e\} \Rightarrow$ asympt. stable.

Barbalat: if $V(\boldsymbol{x}, t)$ lower bounded, $\dot{V} \le 0$, and $\ddot{V}$ bounded $\Rightarrow \lim_{t\to\infty} \dot{V} = 0$.

UUB stability: $\exists T > 0$ dep. on $S, \boldsymbol{x}(t_0)$: $\boldsymbol{x}(t) \in S\ \forall t \ge t_0 + T$.

Linear system $\dot{\boldsymbol{x}} = A\boldsymbol{x}$: asympt.stable $\Leftrightarrow$ GAS $\Leftrightarrow$ exp.stable $\Leftrightarrow$ eig$(A) < 0 \Leftrightarrow \exists P \succ 0 : A^T P + PA = -Q,\ \forall Q \succ 0$.

Linearization: $\Delta\dot{\boldsymbol{x}} = A\Delta\boldsymbol{x}$, $A = \left.\frac{df}{d\boldsymbol{x}}\right|_{\boldsymbol{x}_e}$; eig$(A) < 0 \Rightarrow$ local exp. stability.

3 Regulation in Joint Space

Manipulator model:

$$M(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{c}(\boldsymbol{q}, \dot{\boldsymbol{q}}) + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{\tau} \tag{4}$$

3.1 PD Control

$$\boldsymbol{\tau} = K_P(\boldsymbol{q}_d - \boldsymbol{q}) - K_D\dot{\boldsymbol{q}} \tag{5}$$

GAS (no gravity) via Lyapunov $V = \frac{1}{2}\dot{\boldsymbol{q}}^T M \dot{\boldsymbol{q}} + \frac{1}{2}\boldsymbol{e}^T K_P \boldsymbol{e}$, $\dot{V} = -\dot{\boldsymbol{q}}^T K_D \dot{\boldsymbol{q}} \le 0$. Discrete PD (no tachometer): $\boldsymbol{\tau}_k = K_P \boldsymbol{e}_k + K_D \frac{\boldsymbol{e}_k - \boldsymbol{e}_{k-1}}{T_c}$ (or filtered γ version).

3.2 Gravity Compensation

Cancellation (needs $\boldsymbol{g}(\boldsymbol{q})$ at each step):

$$\boldsymbol{\tau} = K_P(\boldsymbol{q}_d - \boldsymbol{q}) - K_D\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) \tag{6}$$

Compensation (feedforward at $\boldsymbol{q}_d$, cheaper):

$$\boldsymbol{\tau} = K_P(\boldsymbol{q}_d - \boldsymbol{q}) - K_D\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}_d) \tag{7}$$

Condition: $\left\|\frac{\partial \boldsymbol{g}}{\partial \boldsymbol{q}}\right\| \le \alpha$, $K_{P,m} = \sqrt{\lambda_{\min}(K_P^T K_P)}$. If $K_{P,m} > \alpha \Rightarrow$ GAS. Lyapunov: $V = \frac{1}{2}\dot{\boldsymbol{q}}^T M \dot{\boldsymbol{q}} + \frac{1}{2}\boldsymbol{e}^T K_P \boldsymbol{e} + U(\boldsymbol{q}) - U(\boldsymbol{q}_d) + \boldsymbol{e}^T \boldsymbol{g}(\boldsymbol{q}_d)$.

3.3 PID Control

$$\boldsymbol{\tau} = K_P \boldsymbol{e} + K_I \int_{t_0}^{t} \boldsymbol{e}(\sigma)\, d\sigma - K_D \dot{\boldsymbol{q}} \tag{8}$$

1D example stability (Routh): $k_D > 0$, $k_I > 0$, $k_P > \dfrac{m k_I}{k_D}$. Nonlinear PID (saturated integral):

$$\boldsymbol{\tau} = K_P \boldsymbol{e} + K_I \int_{t_0}^{t} \boldsymbol{\phi}(\boldsymbol{q}_d - \boldsymbol{q}(\sigma))\, d\sigma - K_D \dot{\boldsymbol{q}} \tag{9}$$

3.4 Reference Trajectory Regulation

$$\boldsymbol{\tau} = K_P(\boldsymbol{q}_r(t) - \boldsymbol{q}) + K_D(\dot{\boldsymbol{q}}_r(t) - \dot{\boldsymbol{q}}) + \boldsymbol{g}(\boldsymbol{q}) \tag{10}$$

4 Iterative Learning (Regulation)

$$\boldsymbol{\tau} = \gamma K_P \boldsymbol{e} - K_D \dot{\boldsymbol{q}} + \boldsymbol{u}_{i-1} \tag{11}$$

Update: $\boldsymbol{u}_i = \gamma K_P(\boldsymbol{q}_d - \boldsymbol{q}_i) + \boldsymbol{u}_{i-1}$. Convergence: $\lambda_{\min}(K_P) > \alpha$, $\gamma \ge 2 \Rightarrow \boldsymbol{q}_i \to \boldsymbol{q}_d$. Combined (no γ): $\boldsymbol{\tau} = \hat{K}_P \boldsymbol{e} - K_D \dot{\boldsymbol{q}} + \boldsymbol{u}_{i-1}$, condition $\lambda_{\min}(\hat{K}_P) > 2\alpha$.

5 Trajectory Tracking

Feedforward torque: $\boldsymbol{\tau}_d = M(\boldsymbol{q}_d)\ddot{\boldsymbol{q}}_d + \boldsymbol{n}(\boldsymbol{q}_d, \dot{\boldsymbol{q}}_d)$.

1) FF + PD:

$$\boldsymbol{\tau} = \hat{\boldsymbol{\tau}}_d + K_P(\boldsymbol{q}_d - \boldsymbol{q}) + K_D(\dot{\boldsymbol{q}}_d - \dot{\boldsymbol{q}}) \tag{12}$$

(local stability, nominal conditions)

2) Inertia-weighted PD:

$$\boldsymbol{\tau} = \hat{\boldsymbol{\tau}}_d + \hat{M}(\boldsymbol{q}_d)\big(K_P(\boldsymbol{q}_d - \boldsymbol{q}) + K_D(\dot{\boldsymbol{q}}_d - \dot{\boldsymbol{q}})\big) \tag{13}$$

(GAS, nominal conditions)

5.1 Feedback Linearization (FBL)

$$\boldsymbol{\tau} = \hat{M}(\boldsymbol{q})\big(\ddot{\boldsymbol{q}}_d + K_P(\boldsymbol{q}_d - \boldsymbol{q}) + K_D(\dot{\boldsymbol{q}}_d - \dot{\boldsymbol{q}})\big) + \hat{\boldsymbol{n}}(\boldsymbol{q}, \dot{\boldsymbol{q}}) \tag{14}$$

With integral action: add $+K_I \int (\boldsymbol{q}_d - \boldsymbol{q}) dt$ inside parentheses. Nominal closed loop: $\ddot{\boldsymbol{q}} = \boldsymbol{a} = \ddot{\boldsymbol{q}}_d + K_P(\boldsymbol{q}_d - \boldsymbol{q}) + K_D(\dot{\boldsymbol{q}}_d - \dot{\boldsymbol{q}}) \Rightarrow \ddot{\boldsymbol{e}} + K_D \dot{\boldsymbol{e}} + K_P \boldsymbol{e} = 0$ (decoupled, GES).

Per-joint 2nd order: $\ddot{e}_i + k_{Di}\dot{e}_i + k_{Pi} e_i = 0$, $\omega_n = \sqrt{k_{Pi}}$, $\zeta = \dfrac{k_{Di}}{2\sqrt{k_{Pi}}}$.

- Underdamped ($\zeta < 1$): $e_i(t) = e_i(0) e^{-\sigma t}\Big(\cos\omega_d t + \frac{\sigma}{\omega_d}\sin\omega_d t\Big)$, $\sigma = \frac{k_{Di}}{2}$, $\omega_d = \sqrt{\frac{k_{Pi}}{2} - \frac{k_{Di}^2}{4}}$
- Critically damped ($\zeta = 1$): $e_i(t) = e_i(0)(1 + \lambda t)e^{-\lambda t}$, $\lambda = \sqrt{k_{Pi}/2}$
- No overshoot ($\zeta > 1$): $e_i(t) = \frac{e_i(0)}{\lambda_2 - \lambda_1}(\lambda_2 e^{-\lambda_1 t} - \lambda_1 e^{-\lambda_2 t})$
- Overdamped: $e_i(t) = C_1 e^{-\lambda_1 t} + C_2 e^{-\lambda_2 t}$

With integral: Routh conditions $k_{Di} > 0$, $k_{Ii} > 0$, $k_{Di}k_{Pi} - k_{Ii} > 0$. Regulation (FBL, $\dot{\boldsymbol{q}}_d = 0$): $\boldsymbol{\tau} = M(\boldsymbol{q})(K_P(\boldsymbol{q}_d - \boldsymbol{q}) - K_D\dot{\boldsymbol{q}}) + \boldsymbol{n}(\boldsymbol{q}, \dot{\boldsymbol{q}})$ (exponential stability).

5.2 Alternative Trajectory Controller

$$\boldsymbol{\tau} = M(\boldsymbol{q})\ddot{\boldsymbol{q}}_d + C(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}}_d + \boldsymbol{g}(\boldsymbol{q}) + B_v \dot{\boldsymbol{q}}_d + K_P \boldsymbol{e} + K_D \dot{\boldsymbol{e}} \tag{15}$$

GAS (nominal). Regulation reduces to PD+gravity cancellation. Lyapunov: $V = \frac{1}{2}\dot{\boldsymbol{e}}^T M \dot{\boldsymbol{e}} + \frac{1}{2}\boldsymbol{e}^T K_P \boldsymbol{e}$.

5.3 Iterative Trajectory Learning

$$\boldsymbol{\tau} = K_P(\boldsymbol{q}_d - \boldsymbol{q}) + K_D(\dot{\boldsymbol{q}}_d - \dot{\boldsymbol{q}}) + \boldsymbol{v}_k, \quad \boldsymbol{v}_{k+1}(s) = \alpha(s)u'(s) + \beta(s)\boldsymbol{v}_k(s) \tag{16}$$

6 Online Learning (GP-based)

$M = \hat{M} + \Delta M$, $\boldsymbol{n} = \hat{\boldsymbol{n}} + \Delta\boldsymbol{n}$; $\boldsymbol{\tau}_{FBL} = \hat{M}(\boldsymbol{q})\boldsymbol{a} + \hat{\boldsymbol{n}}(\boldsymbol{q}, \dot{\boldsymbol{q}})$

$$\boldsymbol{a}_k = \ddot{\boldsymbol{q}}_{d,k} + K_P(\boldsymbol{q}_{d,k} - \boldsymbol{q}_k) + K_D(\dot{\boldsymbol{q}}_{d,k} - \dot{\boldsymbol{q}}_k) + \boldsymbol{\varepsilon}_k \tag{17}$$

GP regression: $\boldsymbol{x}_k = (\boldsymbol{q}_k, \dot{\boldsymbol{q}}_k, \boldsymbol{u}_k)$, $y_k = \ddot{\boldsymbol{q}}_k - \boldsymbol{u}_k$.

$$\mu(\boldsymbol{x}) = \boldsymbol{k}^T(\boldsymbol{x})(K + \Sigma_\omega)^{-1} Y, \quad \sigma^2(\boldsymbol{x}) = \kappa(\boldsymbol{x}, \boldsymbol{x}) - \boldsymbol{k}^T (K + \Sigma_\omega)^{-1} \boldsymbol{k} \tag{18}$$

7 Task Space Control

7.1 Regulation

$$\boldsymbol{\tau} = J^T(\boldsymbol{q})K_P(\boldsymbol{y}_d - \boldsymbol{y}) - K_D\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) \tag{19}$$

Converges to $A = \{(\boldsymbol{q}, \dot{\boldsymbol{q}}) : \dot{\boldsymbol{q}} = 0 \wedge K_P(\boldsymbol{y}_d - k(\boldsymbol{q})) \in \mathcal{N}(J^T(\boldsymbol{q}))\}$. Variant (task-space damping): $\boldsymbol{\tau} = J^T K_P(\boldsymbol{y}_d - \boldsymbol{y}) - K_D\dot{\boldsymbol{y}} + \boldsymbol{g}(\boldsymbol{q})$.

7.2 FBL in Task Space

$\ddot{\boldsymbol{y}} = J\ddot{\boldsymbol{q}} + \dot{J}\dot{\boldsymbol{q}}$; possible iff $\det J(\boldsymbol{q}) \neq 0$.

$$\boldsymbol{\tau} = MJ^{-1}\boldsymbol{a} + \boldsymbol{c} + \boldsymbol{g} - MJ^{-1}\dot{J}\dot{\boldsymbol{q}} \tag{20}$$

Cartesian wrench form ($m = n$), $\boldsymbol{F} = J^{-T}\boldsymbol{\tau}$:

$$\boldsymbol{F} = \underbrace{(J^{-T}MJ^{-1})}_{M_y}\boldsymbol{a} + \underbrace{J^{-T}(\boldsymbol{c} - MJ^{-1}\dot{J}\dot{\boldsymbol{q}})}_{\boldsymbol{c}_y} + \underbrace{J^{-T}\boldsymbol{g}}_{\boldsymbol{g}_y} \tag{21}$$

Feedforward: $\boldsymbol{a} = \ddot{\boldsymbol{y}}_d + K_D(\dot{\boldsymbol{y}}_d - \dot{\boldsymbol{y}}) + K_P(\boldsymbol{y}_d - \boldsymbol{y})$. Regulation: $\boldsymbol{\tau} = MJ^{-1}(K_P\boldsymbol{e}_y - K_DJ\dot{\boldsymbol{q}}) + \boldsymbol{c} + \boldsymbol{g} - MJ^{-1}\dot{J}\dot{\boldsymbol{q}}$.

Redundant case ($m < n$), weighted pseudoinverse:

$$(JM^{-1})^{\#}_W = W^{-1}M^{-1}J^T(JM^{-1}W^{-1}M^{-1}J^T)^{-1} \tag{22}$$

$$\boldsymbol{\tau} = (JM^{-1})^{\#}_W(\boldsymbol{a} - \dot{J}\dot{\boldsymbol{q}} + JM^{-1}(\boldsymbol{c} + \boldsymbol{g})) + \left(I - (JM^{-1})^{\#}_W JM^{-1}\right)\boldsymbol{\tau}_N \tag{23}$$

Common W: I, M^{-1}, M^{-2}.

8 Adaptive Control

Linear parametrization: $Y(\boldsymbol{q}, \dot{\boldsymbol{q}}, \ddot{\boldsymbol{q}})\boldsymbol{\rho} = \boldsymbol{\tau}$. Non-adaptive FF+PD: $\boldsymbol{\tau} = \hat{M}(\boldsymbol{q}_d)\ddot{\boldsymbol{q}}_d + \hat{C}(\boldsymbol{q}_d, \dot{\boldsymbol{q}}_d)\dot{\boldsymbol{q}}_d + \hat{\boldsymbol{g}}(\boldsymbol{q}_d) + \hat{B}_v\dot{\boldsymbol{q}}_d + K_P\boldsymbol{e} + K_D\dot{\boldsymbol{e}}$.

Modified reference velocity (avoids clamping):

$$\dot{\boldsymbol{q}}_r = \dot{\boldsymbol{q}}_d + \Lambda(\boldsymbol{q}_d - \boldsymbol{q}), \quad \boldsymbol{\sigma} = \dot{\boldsymbol{q}}_r - \dot{\boldsymbol{q}} \tag{24}$$

Adaptive control law:

$$\boldsymbol{\tau} = \hat{M}(\boldsymbol{q})\ddot{\boldsymbol{q}}_r + \hat{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}}_r + \hat{\boldsymbol{g}}(\boldsymbol{q}) + \hat{B}_v\dot{\boldsymbol{q}}_r + K_P\boldsymbol{e} + K_D\dot{\boldsymbol{e}} \tag{25}$$

$$= Y(\boldsymbol{q}, \dot{\boldsymbol{q}}, \dot{\boldsymbol{q}}_r, \ddot{\boldsymbol{q}}_r)\hat{\boldsymbol{\rho}} + K_P\boldsymbol{e} + K_D\dot{\boldsymbol{e}} \tag{26}$$

$$\dot{\hat{\boldsymbol{\rho}}} = \Gamma Y^T(\boldsymbol{q}, \dot{\boldsymbol{q}}, \dot{\boldsymbol{q}}_r, \ddot{\boldsymbol{q}}_r)\,\boldsymbol{\sigma} \tag{27}$$

GAS: $\boldsymbol{e}, \dot{\boldsymbol{e}} \to 0$ (not necessarily $\hat{\boldsymbol{\rho}} \to \boldsymbol{\rho}$, unless persistent excitation). Lyapunov: $V = \frac{1}{2}\boldsymbol{\sigma}^T M\boldsymbol{\sigma} + \frac{1}{2}\boldsymbol{e}^T Q\boldsymbol{e} + \frac{1}{2}\tilde{\boldsymbol{\rho}}^T\Gamma^{-1}\tilde{\boldsymbol{\rho}}$, with $\Lambda = K_D^{-1}K_P$, $Q = 2K_P(I + K_D^{-1}B_v)$.

Adaptive Regulation ($\boldsymbol{q}_d$ const, $\boldsymbol{g}(\boldsymbol{q}) = G(\boldsymbol{q})\boldsymbol{\rho}_g$):

$$\boldsymbol{\tau} = G(\boldsymbol{q})\hat{\boldsymbol{\rho}}_g + K_P(\boldsymbol{q}_d - \boldsymbol{q}) - K_D\dot{\boldsymbol{q}} \tag{28}$$

$$\dot{\hat{\boldsymbol{\rho}}}_g = \gamma G^T(\boldsymbol{q})\left(\frac{2\boldsymbol{e}}{1 + 2\|\boldsymbol{e}\|} - \beta\dot{\boldsymbol{q}}\right) \tag{29}$$

9 Visual Servoing

Pinhole projection: $u = \lambda\frac{X}{Z}$, $v = \lambda\frac{Y}{Z}$ (λ=focal length).

$$\dot{\boldsymbol{p}} = \begin{pmatrix}\dot{u}\\ \dot{v}\end{pmatrix} = \begin{pmatrix}\frac{\lambda}{Z} & 0 & -\frac{u}{Z}\\ 0 & \frac{\lambda}{Z} & -\frac{v}{Z}\end{pmatrix}\dot{\boldsymbol{P}} = J_1(u, v, \lambda)\dot{\boldsymbol{P}} \tag{30}$$

$$\dot{\boldsymbol{P}} = \begin{pmatrix}-1 & 0 & 0 & 0 & -Z & Y\\ 0 & -1 & 0 & Z & 0 & -X\\ 0 & 0 & -1 & -Y & X & 0\end{pmatrix}\begin{pmatrix}\boldsymbol{V}\\ \boldsymbol{\Omega}\end{pmatrix} = J_2(X, Y, Z)\begin{pmatrix}\boldsymbol{V}\\ \boldsymbol{\Omega}\end{pmatrix} \tag{31}$$

Interaction matrix J_p (point feature):

$$\dot{\boldsymbol{p}} = \underbrace{\begin{pmatrix}-\frac{\lambda}{Z} & 0 & \frac{u}{Z} & \frac{uv}{\lambda} & -(\lambda + \frac{u^2}{\lambda}) & v\\ 0 & -\frac{\lambda}{Z} & \frac{v}{Z} & \lambda + \frac{v^2}{\lambda} & -\frac{uv}{\lambda} & -u\end{pmatrix}}_{J_p}\begin{pmatrix}\boldsymbol{V}\\ \boldsymbol{\Omega}\end{pmatrix} \tag{32}$$

Image Jacobian: $\dot{\boldsymbol{f}} = J_p(\boldsymbol{f}, Z)J(\boldsymbol{q})\dot{\boldsymbol{q}} = J_I(\boldsymbol{f}, Z, \boldsymbol{q})\dot{\boldsymbol{q}}$. Control law: $\dot{\boldsymbol{q}} = J_I^{\#}(\dot{\boldsymbol{f}}_d + K(\boldsymbol{f}_d - \boldsymbol{f})) + (I - J_I^{\#}J_I)\dot{\boldsymbol{q}}_N$. Error dynamics: $\dot{\boldsymbol{e}} = -J_I\hat{J}_I^{\#}K\boldsymbol{e}$; local convergence if $J_I\hat{J}_I^{\#} \succ 0$. Depth strategies: constant $Z \approx Z^*$; online EKF/observer; model-based PnP.

Task Sequencing (IBVS):

$$\boldsymbol{u} = J_1^{\#}k_1\boldsymbol{e}_1 + (I - J_1^{\#}J_1)\boldsymbol{u}_0, \quad \boldsymbol{u} = (I - J_1^{\#}J_1)J_2^T k_2\boldsymbol{e}_2 \tag{33}$$

10 Interaction with Environment

10.1 Constrained Dynamics

Constraint: $\boldsymbol{h}(\boldsymbol{q}) = \boldsymbol{f}(k(\boldsymbol{q})) = \boldsymbol{0}$; Augmented Lagrangian:

$$\mathcal{L}_a = T(\boldsymbol{q}, \dot{\boldsymbol{q}}) - U(\boldsymbol{q}) + \boldsymbol{\lambda}^T\boldsymbol{h}(\boldsymbol{q}) \tag{34}$$

$$M(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{c} + \boldsymbol{g} = \boldsymbol{\tau} + A^T(\boldsymbol{q})\boldsymbol{\lambda}, \quad \boldsymbol{h}(\boldsymbol{q}) = \boldsymbol{0}, \ A = \frac{\partial\boldsymbol{h}}{\partial\boldsymbol{q}} \tag{35}$$

Eliminating multipliers: $\boldsymbol{\lambda} = (A_M^{\#})^T(\boldsymbol{c} + \boldsymbol{g} - \boldsymbol{\tau}) - (AM^{-1}A^T)^{-1}\dot{A}\dot{\boldsymbol{q}}$, with

$$A_M^{\#} = M^{-1}A^T(AM^{-1}A^T)^{-1} \tag{36}$$

Final constrained model:

$$M\ddot{\boldsymbol{q}} = (I - A^T(A_M^{\#})^T)(\boldsymbol{\tau} - \boldsymbol{c} - \boldsymbol{g}) - MA_M^{\#}\dot{A}\dot{\boldsymbol{q}} \tag{37}$$

10.2 Reduced Dynamics

Choose $D(\boldsymbol{q})$: $\begin{pmatrix}A\\ D\end{pmatrix}$ nonsingular, $\begin{pmatrix}A\\ D\end{pmatrix}^{-1} = (E \ \ G)$. $\boldsymbol{u} = D\dot{\boldsymbol{q}}$ (reduced velocity); $\dot{\boldsymbol{q}} = G\boldsymbol{u}$.

$$(G^TMG)\dot{\boldsymbol{u}} = G^T(\boldsymbol{\tau} - \boldsymbol{c} - \boldsymbol{g} + M(E\dot{A} + G\dot{D})G\boldsymbol{u}) \tag{38}$$

$$\boldsymbol{\lambda} = E^T(MG\dot{\boldsymbol{u}} - M(E\dot{A} + G\dot{D})G\boldsymbol{u} + \boldsymbol{c} + \boldsymbol{g} - \boldsymbol{\tau}) \tag{39}$$

FBL for reduced model:

$$\boldsymbol{\tau} = (\boldsymbol{c} + \boldsymbol{g} - M(E\dot{A} + G\dot{D})G\boldsymbol{u}) + MG\boldsymbol{a} - A^T\boldsymbol{f} \tag{40}$$

gives $\dot{\boldsymbol{u}} = \boldsymbol{a}$, $\boldsymbol{\lambda} = \boldsymbol{f}$. $\boldsymbol{a} = \dot{\boldsymbol{u}}_d + K_P(\boldsymbol{u}_d - \boldsymbol{u}) + K_I\int(\boldsymbol{u}_d - \boldsymbol{u})$.

10.3 Compliance

Combined stiffness: $K = \dfrac{K_sK_e}{K_s + K_e}$. Force control loop (1-mass): $m_r\ddot{x}_r = F - k_sx_r - (b_r + b_s)\dot{x}_r$, $F_c = k_sx_r$, $F = k_f(F_d - F_c)$:

$$W(s) = \frac{F_c(s)}{F_d(s)} = \frac{k_sk_f}{m_rs^2 + (b_r + b_s)s + k_s(1 + k_f)} \tag{41}$$

Steady-state: $\lim_{s\to 0} W(s) = \dfrac{k_f}{1 + k_f} \neq 1$.

10.4 Impedance Control

Task model: $M_y\ddot{\boldsymbol{y}} + \boldsymbol{c}_y + \boldsymbol{g}_y = J^{-T}\boldsymbol{\tau} + \boldsymbol{F}_y$. FBL: $\boldsymbol{\tau} = J^T(M_y\boldsymbol{a} + \boldsymbol{c}_y + \boldsymbol{g}_y - \boldsymbol{F}_y) \Rightarrow \ddot{\boldsymbol{y}} = \boldsymbol{a}$. Target impedance: $M_m(\ddot{\boldsymbol{y}} - \ddot{\boldsymbol{y}}_d) + B_m(\dot{\boldsymbol{y}} - \dot{\boldsymbol{y}}_d) + K_m(\boldsymbol{y} - \boldsymbol{y}_d) = \boldsymbol{F}_y$.

$$\boldsymbol{a} = \ddot{\boldsymbol{y}}_d + M_m^{-1}\left(B_m(\dot{\boldsymbol{y}}_d - \dot{\boldsymbol{y}}) + K_m(\boldsymbol{y}_d - \boldsymbol{y}) + \boldsymbol{F}_y\right) \tag{42}$$

Joint space (with force meas.):

$$\boldsymbol{\tau} = MJ^{-1}\left(\ddot{\boldsymbol{y}}_d - \dot{J}\dot{\boldsymbol{q}} + M_m^{-1}(B_m(\dot{\boldsymbol{y}}_d - \dot{\boldsymbol{y}}) + K_m(\boldsymbol{y}_d - \boldsymbol{y}))\right) + \boldsymbol{c} + \boldsymbol{g} + J^T(M_yM_m^{-1} - I)\boldsymbol{F}_y \tag{43}$$

Without force meas. ($M_m = M_y$):

$$\boldsymbol{\tau} = MJ^{-1}(\ddot{\boldsymbol{y}}_d - \dot{J}\dot{\boldsymbol{q}}) + \boldsymbol{c} + \boldsymbol{g} + J^T(B_m(\dot{\boldsymbol{y}}_d - \dot{\boldsymbol{y}}) + K_m(\boldsymbol{y}_d - \boldsymbol{y})) \tag{44}$$

Regulation: $\boldsymbol{\tau} = \boldsymbol{g} + J^T(K_m(\boldsymbol{y}_d - \boldsymbol{y}) - B_m\dot{\boldsymbol{y}})$ (PD+gravity cancellation). Elastic environment: $\boldsymbol{F}_y = K_e(\boldsymbol{y}_e - \boldsymbol{y})$; equilibrium $\boldsymbol{y}_E = (K_m + K_e)^{-1}(K_m\boldsymbol{y}_d + K_e\boldsymbol{y}_e)$.

10.5 Admittance Control

$$\boldsymbol{\tau}_c = J^T\boldsymbol{F}_c \Rightarrow \dot{\boldsymbol{q}} = C_qJ^T\boldsymbol{F}_c \quad \text{(joint space)}; \qquad \dot{\boldsymbol{q}} = J^{-1}C_y\boldsymbol{F}_c \text{ (task space)} \tag{45}$$

11 Hybrid Control

Selector matrices: $\boldsymbol{t}_e = S_t(\boldsymbol{q})\boldsymbol{\nu},\ \boldsymbol{\nu} \in \mathbb{R}^{6-k}$; $\boldsymbol{w}_{env} = S_w(\boldsymbol{q})\boldsymbol{\lambda},\ \boldsymbol{\lambda} \in \mathbb{R}^k$; $S_w^T S_t = O$.

Force/Velocity Control (FBL):

$$\boldsymbol{\tau} = MJ^{-1}(S_t\boldsymbol{a}_\nu + \dot{S}_t\boldsymbol{\nu} - \dot{J}\dot{\boldsymbol{q}}) + J^T S_w \boldsymbol{a}_\lambda + \boldsymbol{c} + \boldsymbol{g} \tag{46}$$

gives $\dot{\boldsymbol{\nu}} = \boldsymbol{a}_\nu,\ \boldsymbol{\lambda} = \boldsymbol{a}_\lambda$.

$$\boldsymbol{a}_\lambda = \boldsymbol{\lambda}_d + K_{I\lambda}\int(\boldsymbol{\lambda}_d - \boldsymbol{\lambda})\,dt \tag{47}$$

$$\boldsymbol{a}_\nu = \dot{\boldsymbol{\nu}}_d + K_{P\nu}(\boldsymbol{\nu}_d - \boldsymbol{\nu}) + K_{I\nu}\int(\boldsymbol{\nu}_d - \boldsymbol{\nu})\,dt \tag{48}$$

Free-motion variable via $r(\boldsymbol{q})$: $\boldsymbol{a}_\nu = \ddot{r}_d + K_{Dr}(\dot{r}_d - \dot{r}) + K_{Pr}(r_d - r)$. Force measurement: $\boldsymbol{\lambda} = S_w^{\#}\boldsymbol{w}_{env}$.

12 Fault Detection

FDI Steps: Detection → Isolation → Identification → Accommodation. Faulty dynamic model:

$$M\ddot{\boldsymbol{q}} + \boldsymbol{c} + \boldsymbol{g} + B_v\dot{\boldsymbol{q}} = \boldsymbol{\tau} - \boldsymbol{\tau}_f \tag{49}$$

Fault types: total ($\tau_{f,i} = \tau_i$), partial ($\varepsilon\tau_i$), saturation, bias, block ($\ddot{q}_i = -\lambda_i\dot{q}_i$).

12.1 Residual (Momentum-Based) Method

Generalized momentum: $\boldsymbol{p} = M\dot{\boldsymbol{q}}$.

$$\dot{p}_i = \tau_i - \tau_{f,i} - \alpha_i(\boldsymbol{q},\dot{\boldsymbol{q}}), \quad \alpha_i = -\tfrac{1}{2}\dot{\boldsymbol{q}}^T\frac{\partial M}{\partial q_i}\dot{\boldsymbol{q}} + g_i + b_{vi}\dot{q}_i \tag{50}$$

$$\boldsymbol{r} = K\left[\int(\boldsymbol{\tau} - \boldsymbol{\alpha}(\boldsymbol{q},\dot{\boldsymbol{q}}) - \boldsymbol{r})\,dt - \boldsymbol{p}\right] \tag{51}$$

Residual dynamics: $\dot{\boldsymbol{r}} = K(\boldsymbol{\tau}_f - \boldsymbol{r})$, per-joint $\dfrac{r_i}{\tau_{f,i}} = \dfrac{1}{1+s/K_i}$. Threshold filtering: $r_{i,filtered} = r_i$ if $|r_i| > \epsilon$, else 0.

Force-sensor fault isolation (adjoint Jacobian, avoids singularities):

$$A = {}^2J_{adj} = (\det{}^2J)({}^2J)^{-1}, \quad \boldsymbol{p}^* = A^T M\dot{\boldsymbol{q}} \tag{52}$$

$$\boldsymbol{r} = A^T M\dot{\boldsymbol{q}} - \boldsymbol{\zeta}, \quad \dot{\boldsymbol{\zeta}} = \dot{\boldsymbol{p}}^* + K[A^T M\dot{\boldsymbol{q}} - \boldsymbol{\zeta}] \tag{53}$$

12.2 Collision Detection

$$M\ddot{\boldsymbol{q}} + \boldsymbol{c} + \boldsymbol{g} = \boldsymbol{\tau} + \boldsymbol{\tau}_c, \quad \boldsymbol{\tau}_c = J^T\boldsymbol{F}_c \tag{54}$$

$$\boldsymbol{r} = K\left(\boldsymbol{p} - \int(\boldsymbol{\tau} + \boldsymbol{c} - \boldsymbol{g} + \boldsymbol{r})\,dt - \boldsymbol{p}(0)\right) \tag{55}$$

$$\dot{\boldsymbol{r}} = K(\boldsymbol{\tau}_c - \boldsymbol{r}) \ \Rightarrow\ \boldsymbol{r} \simeq \boldsymbol{\tau}_c = J^T\boldsymbol{F}_c \tag{56}$$

13 Matrix/Math Appendix

Matrix 2-norm (max singular value): $\|A\|_2 = \sigma_{\max}(A) = \sqrt{\lambda_{\max}(A^TA)}$; for symmetric PD M: $\|M\|_2 = \lambda_{\max}(M)$. Frobenius bound: $\|A\|_2 \le \|A\|_F = \sqrt{\sum_{i,j}|A_{ij}|^2}$.

2×2 inverse: $A^{-1} = \dfrac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.

3×3 inverse: $A^{-1} = \dfrac{1}{\det A}C^T$, cofactor $c_{ij} = (-1)^{i+j}m_{ij}$.

Derivative of quadratic form: $\frac{d}{dt}(\boldsymbol{x}^TA\boldsymbol{x} + \boldsymbol{c}^T\boldsymbol{x}) = \boldsymbol{x}^T\dot{A}\boldsymbol{x} + 2\boldsymbol{x}^TA\dot{\boldsymbol{x}} + \boldsymbol{c}^T\dot{\boldsymbol{x}}$.

Angle between vectors: $\theta = \arccos\left(\dfrac{\boldsymbol{a}\cdot\boldsymbol{b}}{\|\boldsymbol{a}\|\|\boldsymbol{b}\|}\right)$.

Trig sum/difference:

$$\sin(x \pm y) = \sin x\cos y \pm \cos x\sin y \tag{57}$$

$$\cos(x \pm y) = \cos x\cos y \mp \sin x\sin y \tag{58}$$

$$\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x\tan y} \tag{59}$$